

# SecureScan - AI-Powered Fraud Detection and Cybersecurity Mobile Application



**Shresth Mantry (2201187)**

Advisor: Dr. Angshuman Jana

Department of Computer Science and Engineering  
Indian Institute of Information Technology Guwahati

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# Contents

|                                           |           |
|-------------------------------------------|-----------|
| <b>Contents</b>                           | <b>1</b>  |
| <b>1 Introduction</b>                     | <b>2</b>  |
| <b>2 App Purpose</b>                      | <b>2</b>  |
| <b>3 Target Audience</b>                  | <b>3</b>  |
| <b>4 Core Features</b>                    | <b>3</b>  |
| 4.1 AI-Powered Threat Detection . . . . . | 3         |
| 4.1.1 QR Code Scanner . . . . .           | 4         |
| 4.1.2 Link Checker . . . . .              | 4         |
| 4.2 User Features . . . . .               | 4         |
| 4.2.1 AI Security Assistant . . . . .     | 5         |
| 4.2.2 Community Platform . . . . .        | 5         |
| 4.2.3 Authentication System . . . . .     | 6         |
| 4.2.4 Email Verification (OTP) . . . . .  | 6         |
| 4.2.5 Profile & Statistics . . . . .      | 7         |
| 4.3 Machine Learning Backend . . . . .    | 7         |
| 4.3.1 Flask ML Service . . . . .          | 7         |
| 4.4 Admin Features . . . . .              | 8         |
| 4.4.1 Admin Dashboard . . . . .           | 8         |
| <b>5 Technologies Used</b>                | <b>8</b>  |
| <b>6 System Architecture</b>              | <b>9</b>  |
| <b>7 Security Implementation</b>          | <b>10</b> |
| <b>8 Future Scope</b>                     | <b>11</b> |
| <b>9 Conclusion</b>                       | <b>12</b> |
| <b>10 Screenshots</b>                     | <b>13</b> |
| <b>11 Appendix</b>                        | <b>19</b> |
| 11.1 Repository Information . . . . .     | 19        |

# 1 Introduction

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## Overview

**SecureScan** is a comprehensive cybersecurity and fraud detection mobile application designed to protect users from online threats, phishing attacks, QR code fraud, and malicious URLs. The app leverages artificial intelligence and machine learning to provide real-time threat detection and analysis, making cybersecurity accessible to everyone.

In today's digital age, cyber threats have become increasingly sophisticated, targeting users through various vectors including malicious QR codes, phishing URLs, and social engineering attacks. SecureScan addresses this critical need by combining advanced machine learning models with an intuitive mobile interface, enabling users to verify the safety of digital content before interaction.

The application integrates multiple security features including real-time QR code analysis, URL verification, an AI-powered cybersecurity chatbot, and a community platform for sharing threat intelligence. By providing instant threat assessment with confidence scores and detailed security reports, SecureScan empowers users to make informed decisions about their digital interactions.

Built with modern technologies including React Native for cross-platform compatibility, Flask-based ML services for intelligent threat detection, and MongoDB for robust data management, SecureScan represents a holistic approach to mobile cybersecurity that prioritizes both security and user experience.

## 2 App Purpose

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SecureScan addresses the growing concern of digital fraud and cyber threats in today's interconnected world. With the increasing sophistication of phishing attacks, malicious QR codes, and fraudulent websites, users need a reliable tool to verify the safety of digital content before interaction. The application serves to:

- Protect users from phishing attacks and malicious URLs
  - Detect and analyze potentially dangerous QR codes
  - Provide real-time fraud detection using AI/ML models
  - Educate users about cybersecurity threats through an AI assistant
  - Create a community platform for sharing security insights
  - Track and monitor security scans for user awareness
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The app's AI-powered detection system provides instant analysis of potential threats, helping users make informed decisions about their digital interactions.

### 3 Target Audience

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SecureScan targets a diverse audience of individuals and organizations concerned with digital security, including:

- General smartphone users who want protection from online fraud
- Online shoppers who encounter suspicious links
- Social media users exposed to phishing attempts
- Elderly users vulnerable to digital scams
- Students and professionals sharing sensitive information online
- Organizations seeking mobile security solutions for employees

The app is particularly valuable for users who frequently interact with QR codes, click on shared links, or conduct financial transactions online.

## 4 Core Features

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### 4.1 AI-Powered Threat Detection

The core functionality of SecureScan revolves around intelligent threat detection using machine learning models. These features enable users to scan and verify potentially malicious content before interaction.

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### 4.1.1 QR Code Scanner

#### QR Code Scanner

The QR Code Scanner uses the device camera to analyze QR codes in real-time, identifying fraudulent or malicious codes through ML-powered detection.

- **Real-time scanning:** Instantly analyze QR codes using device camera
- **ML fraud detection:** Machine learning models classify threats with confidence scores
- **Threat classification:** Categorizes codes as Phishing, Malware, Defacement, or Benign
- **Image upload support:** Analyze QR codes from saved images for offline verification
- **Detailed reports:** Provides threat information and safety recommendations
- **History tracking:** Automatically maintains scan history for future reference

### 4.1.2 Link Checker

#### Link Checker

The Link Checker validates URLs before users click on them, using AI to detect phishing attempts and malicious websites.

- **URL validation:** Analyzes web addresses for malicious patterns and fraud indicators
- **AI-powered detection:** Identifies phishing sites and dangerous links using intelligent algorithms
- **Complex URL support:** Handles URLs with query parameters and encoded characters
- **Real-time analysis:** Provides instant security assessment of web links
- **Visual indicators:** Color-coded threat levels for quick understanding
- **Security reports:** Detailed analysis explaining why a link is flagged as dangerous

## 4.2 User Features

Beyond threat detection, SecureScan provides comprehensive user engagement features including AI assistance, community interaction, and robust security measures.

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#### 4.2.1 AI Security Assistant

##### AI Assistant

An intelligent chatbot powered by Google Gemini Pro that provides personalized cybersecurity guidance and answers security-related questions.

- **Gemini Pro integration:** Leverages Google's advanced AI for accurate responses
- **Context-aware chatbot:** Understands cybersecurity context and provides relevant advice
- **Specialized knowledge:** Trained on fraud detection and security best practices
- **Natural explanations:** Explains threats in simple, understandable language
- **Interactive Q&A:** Ask any security-related question and get instant answers
- **History management:** Maintains conversation history for reference

#### 4.2.2 Community Platform

##### Community Platform

A social platform where users can share security experiences, discuss threats, and learn from each other's encounters with fraud.

- **Share insights:** Post security experiences and fraud encounters to help others
- **Rich content creation:** Create detailed posts with descriptions and context
- **Discussion system:** Comment on posts to engage in security conversations
- **Engagement features:** Like posts and comments to show appreciation
- **Admin moderation:** Designated admin ensures content quality and safety
- **User tracking:** Monitor your contributions and community interactions

### 4.2.3 Authentication System

#### Authentication System

A secure authentication system that protects user accounts with industry-standard encryption and token-based session management.

- **Secure registration:** Email validation ensures legitimate user accounts
- **OTP verification:** Two-step verification through email for added security
- **JWT sessions:** Token-based authentication for secure API access
- **Password encryption:** bcrypt hashing protects user passwords
- **Secure storage:** Tokens stored safely using AsyncStorage
- **Profile management:** Users can update their account information securely

### 4.2.4 Email Verification (OTP)

#### Email Verification

A seamless OTP-based email verification system that confirms user identity during registration with branded, secure emails.

- **6-digit OTP:** Simple yet secure one-time password sent to email
- **Branded emails:** Professional HTML templates with gradient design
- **Security timeout:** 5-minute expiration prevents unauthorized access
- **Auto-verification:** Automatically verifies when all digits are entered
- **Resend functionality:** Request new OTP with cooldown to prevent abuse
- **Enhanced UX:** Auto-focus inputs and shake animation for errors

#### 4.2.5 Profile & Statistics

##### Profile & Statistics

A comprehensive profile dashboard that tracks user activity, scan history, and security statistics for personal awareness and progress monitoring.

- **Profile management:** View and update personal account information
- **Scan statistics:** Real-time counters for QR scans, URL checks, and threats detected
- **History tracking:** AsyncStorage maintains persistent scan history locally
- **Account details:** Display account creation date and membership duration
- **Activity metrics:** Track community posts and engagement statistics
- **Analytics:** Visual indicators show security activity trends

### 4.3 Machine Learning Backend

The technical foundation that powers SecureScan's intelligent threat detection, built with Python and Flask for efficient model serving.

#### 4.3.1 Flask ML Service

##### ML Service

A Python-based microservice that hosts trained machine learning models and provides RESTful APIs for fraud detection.

- **Flask API:** Lightweight Python web framework serves ML models efficiently
- **BERT-based model:** Pre-trained bert-base-uncased model with 95% accuracy for URL threat classification
- **REST endpoints:** Standard HTTP APIs for QR and URL analysis
- **Image processing:** OpenCV-based computer vision to detect and decode QR codes
- **Pattern analysis:** URL structure and behavior analysis for threat detection
- **Confidence scoring:** Probabilistic scoring indicates detection certainty



4.4 Admin Features

Special administrative capabilities for content moderation and community management to maintain platform safety and quality.

4.4.1 Admin Dashboard

Admin Dashboard

Elevated privileges for the designated administrator to moderate content and manage the community platform effectively.

- **Admin account:** Single designated admin (shresthmantry72003@gmail.com)
- **Content moderation:** Delete any inappropriate posts or comments
- **Admin badge:** Visual indicator distinguishes admin from regular users
- **Enhanced privileges:** Special permissions for security and content management
- **Community oversight:** Monitor and maintain platform quality and safety

5 Technologies Used

| Technology Stack  |                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
| Frontend          | React Native 0.74.5, Expo SDK 51, TypeScript, Expo Router                                                           |
| Backend           | Node.js, Express 4.18.2, MongoDB Atlas, JWT Authentication, Nodemailer 6.9.7, bcryptjs                              |
| ML Service        | Python 3.11, Flask 3.0.0, TensorFlow/PyTorch, OpenCV, NumPy, Scikit-learn                                           |
| AI Integration    | Google Gemini Pro API, @google/generative-ai                                                                        |
| Storage           | MongoDB for user data and posts, AsyncStorage for local data, In-memory OTP storage (production: Redis recommended) |
| UI Libraries      | expo-linear-gradient, expo-blur, react-native-reanimated, @expo/vector-icons (Ionicons)                             |
| Development Tools | Git, GitHub, VS Code, Expo CLI, npm, Postman (API testing)                                                          |

## 6 System Architecture

### Architecture Overview

SecureScan follows a modern three-tier architecture:

#### 1. Mobile Frontend (React Native + Expo)

- Cross-platform iOS and Android support
- TypeScript for type safety
- Expo Router for navigation
- Context API for state management
- AsyncStorage for local persistence

#### 2. Backend API (Node.js + Express)

- RESTful API architecture
- JWT authentication middleware
- MongoDB for data persistence
- Nodemailer for email services
- CORS-enabled for cross-origin requests

#### 3. ML Service (Python + Flask)

- Separate microservice for ML models
- Flask REST API for model inference
- Image processing pipeline
- URL pattern analysis
- Threat classification engine

#### 4. External Services

- Google Gemini Pro for AI chat
- Gmail SMTP for OTP emails

## 7 Security Implementation

### Security Measures

#### Authentication & Authorization

- **JWT tokens** - Stateless authentication for secure session management without server-side storage
- **Password hashing (bcrypt)** - 10-round one-way encryption makes passwords unrecoverable
- **Email verification with OTP** - Two-factor authentication ensures account authenticity
- **Secure token storage** - AsyncStorage encrypts and protects tokens on device
- **Role-based access control** - Separates user and admin permissions for content management

#### Data Protection

- **HTTPS communications** - SSL/TLS encryption prevents man-in-the-middle attacks
- **Environment variables** - Keeps sensitive credentials separate from source code
- **MongoDB encryption** - Protects database credentials and user data access
- **Gmail App Passwords** - Application-specific passwords reduce security risks
- **Input validation** - Prevents SQL injection, XSS, and malicious code execution

#### OTP Security

- **5-minute expiration** - Minimizes window for interception or brute-force attacks
- **One-time use** - Tokens invalid after single use, prevents replay attacks
- **Rate limiting** - Restricts OTP request frequency to prevent spam
- **Secure delivery** - Gmail's encrypted infrastructure ensures safe transmission
- **Auto-deletion** - Removes OTPs immediately after verification

## 8 Future Scope

### Future Development Plans

#### Enhanced ML Capabilities

- Implement deep learning models for improved accuracy
- Add real-time phishing website detection and SMS spam detection
- Email phishing analysis and file malware scanning

#### Advanced Security Features

- Two-factor authentication (2FA) and biometric authentication (fingerprint/face ID)
- VPN integration for secure browsing
- Dark web monitoring for leaked credentials

#### AI & Automation

- Automated threat alerts and AI-powered security recommendations
- Predictive threat analysis based on user behavior
- Voice-activated security assistant

#### Analytics & Reporting

- Detailed security analytics dashboard with weekly/monthly reports
- Trend analysis and geographical threat mapping
- Export capabilities and SIEM system integration

#### Enterprise Features

- Organization accounts with multi-user support and admin dashboard
- API access for third-party integrations with custom branding
- Compliance reporting (GDPR, HIPAA) and bulk scanning capabilities

## 9 Conclusion

### Project Summary

**SecureScan** represents a comprehensive approach to mobile cybersecurity, combining cutting-edge machine learning, artificial intelligence, and modern mobile development practices. Developed entirely by **Shresth Mantry**, the application demonstrates a full-stack implementation of a real-world security solution.

The project successfully integrates multiple complex technologies including React Native for cross-platform mobile development, Node.js for backend services, Python Flask for ML model serving, MongoDB for data persistence, and Google Gemini Pro for AI-powered assistance. The implementation of email OTP verification adds an extra layer of security, while the community platform fosters user engagement and knowledge sharing.

Key achievements of this project include:

- Functional AI-powered fraud detection system
- Real-time QR code and URL threat analysis
- Secure authentication with email verification
- Interactive AI security assistant
- Community platform for security awareness
- Admin moderation system
- Modern, user-friendly interface
- Comprehensive documentation

The modular architecture allows for easy maintenance and feature expansion, with a clear roadmap for future development. As cyber threats continue to evolve, **SecureScan** is positioned to adapt and provide users with cutting-edge protection.

This project demonstrates proficiency in full-stack development, machine learning integration, API design, mobile UI/UX, security best practices, and project management - all essential skills for modern software engineering.

## 10 Screenshots

### Application Screenshots

The following screenshots showcase the key features and interfaces of the SecureScan application.

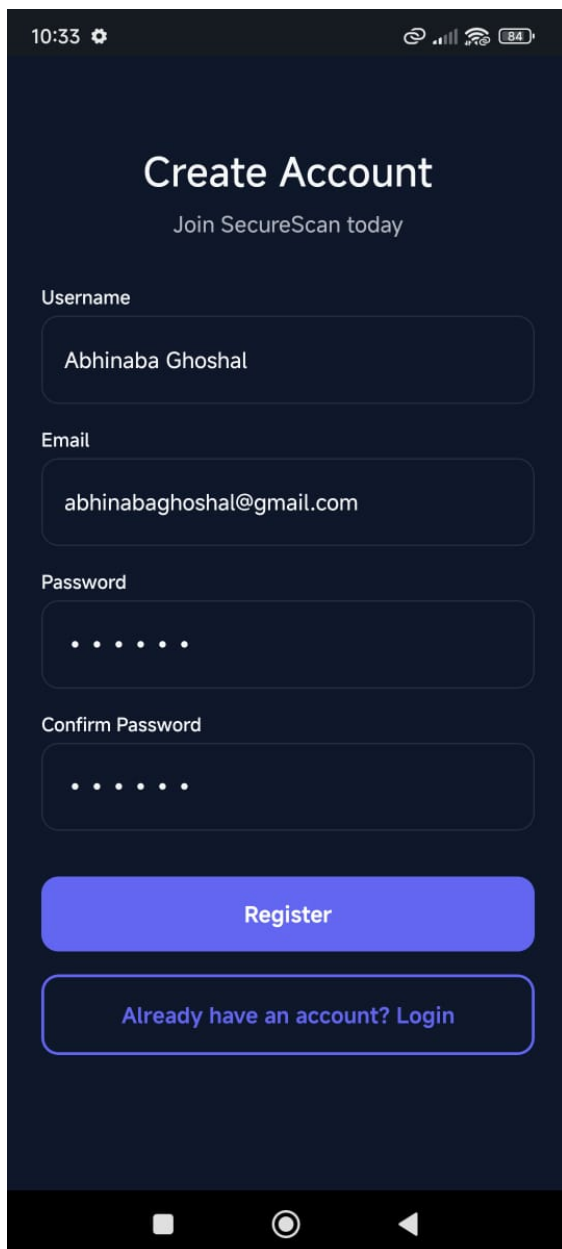


Figure 1: Register Screen

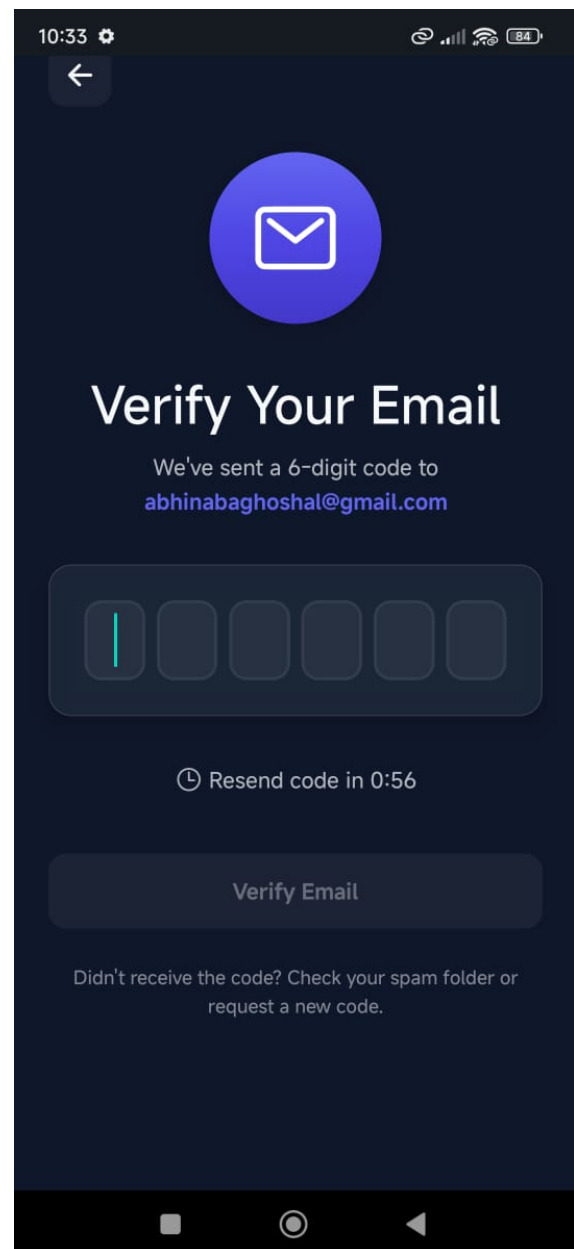


Figure 2: OTP Verification Screen

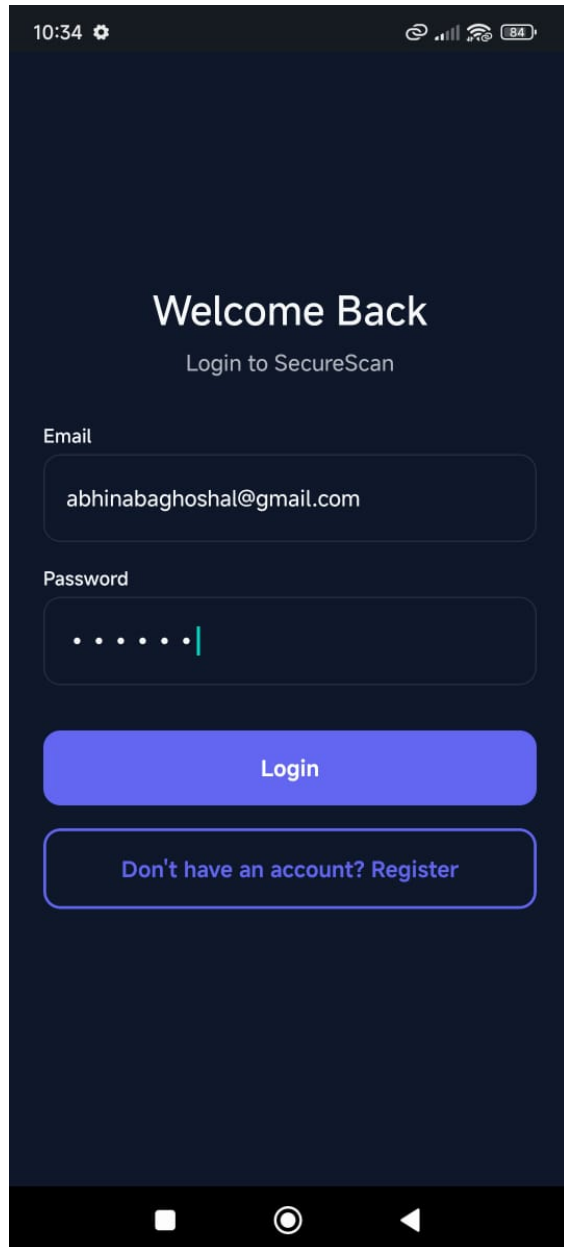


Figure 3: Login Screen

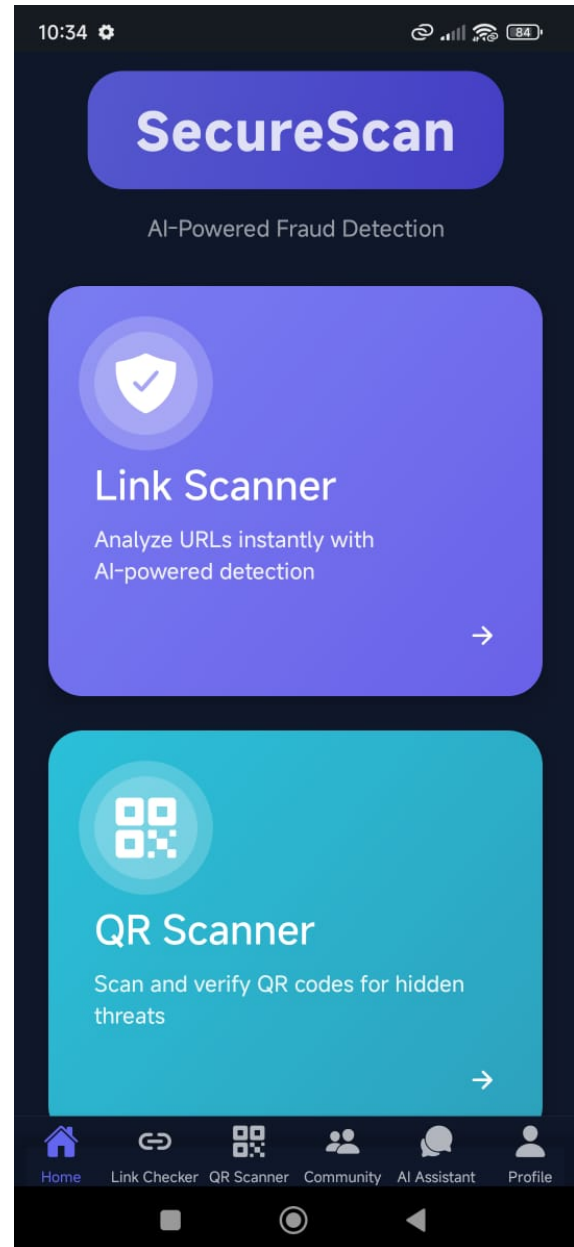


Figure 4: Home Dashboard

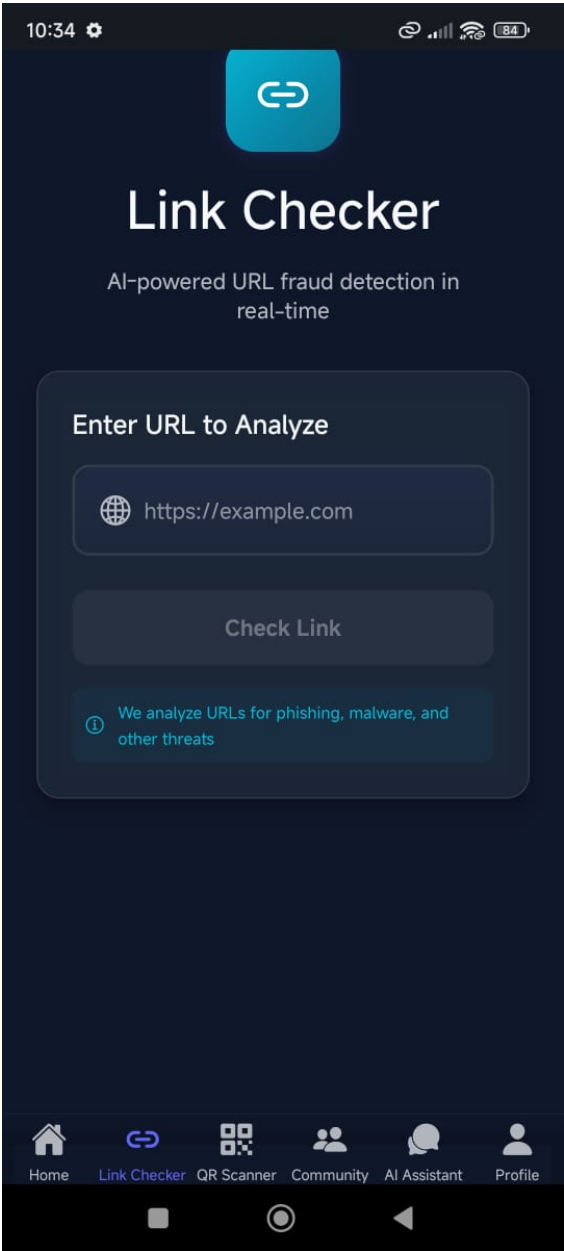


Figure 5: Link Checker Interface

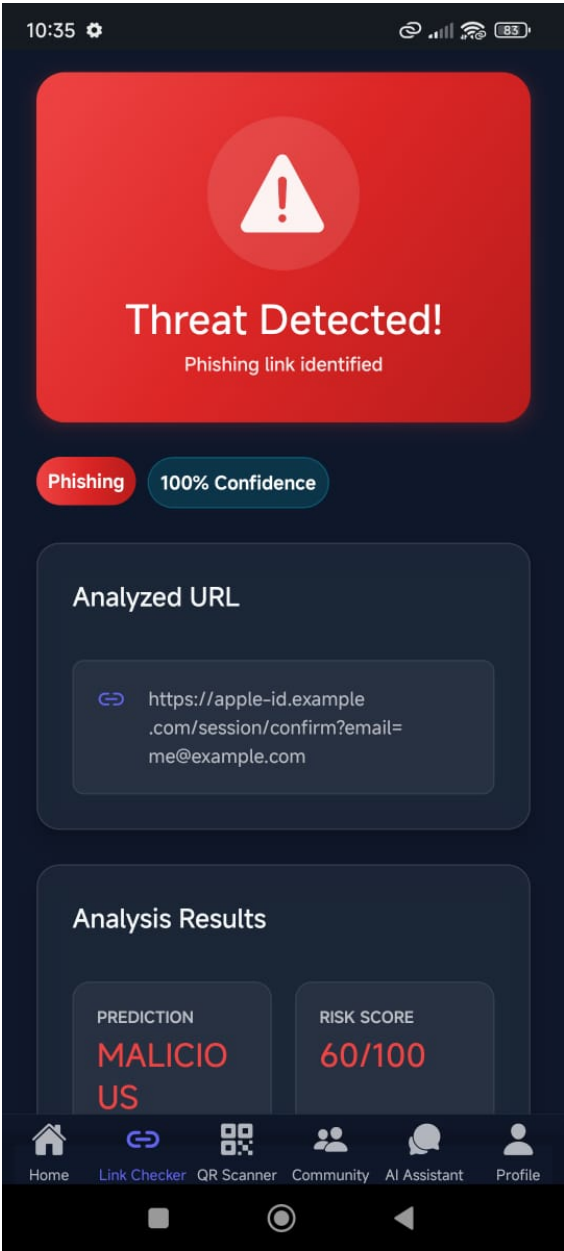


Figure 6: Threat Detection Results



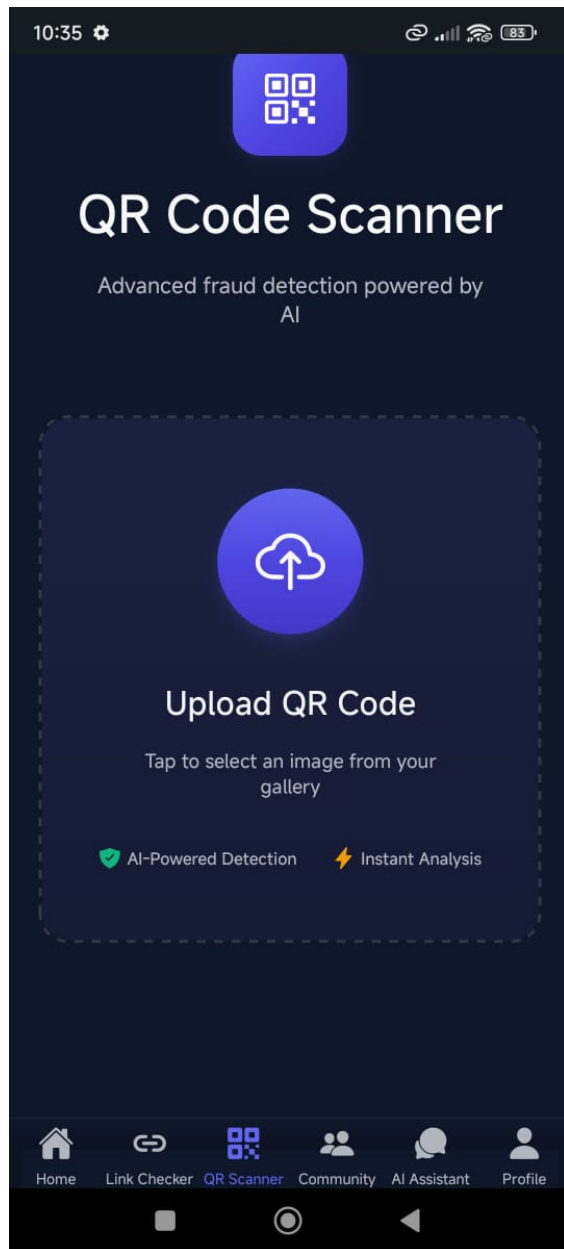


Figure 7: QR Code Scanner

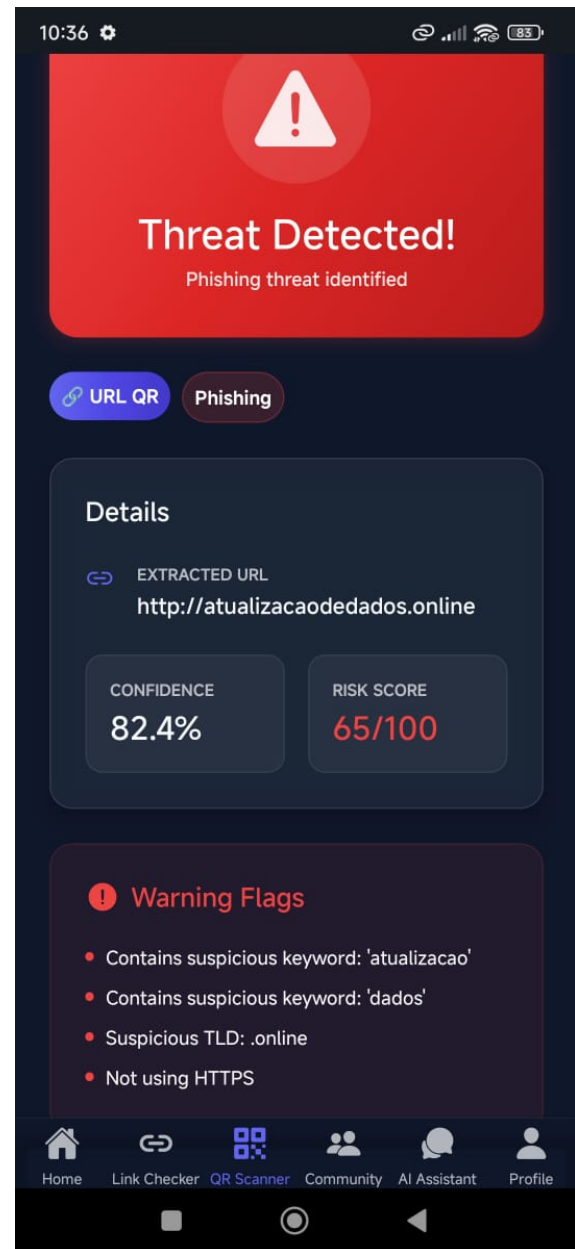


Figure 8: QR Scan Results

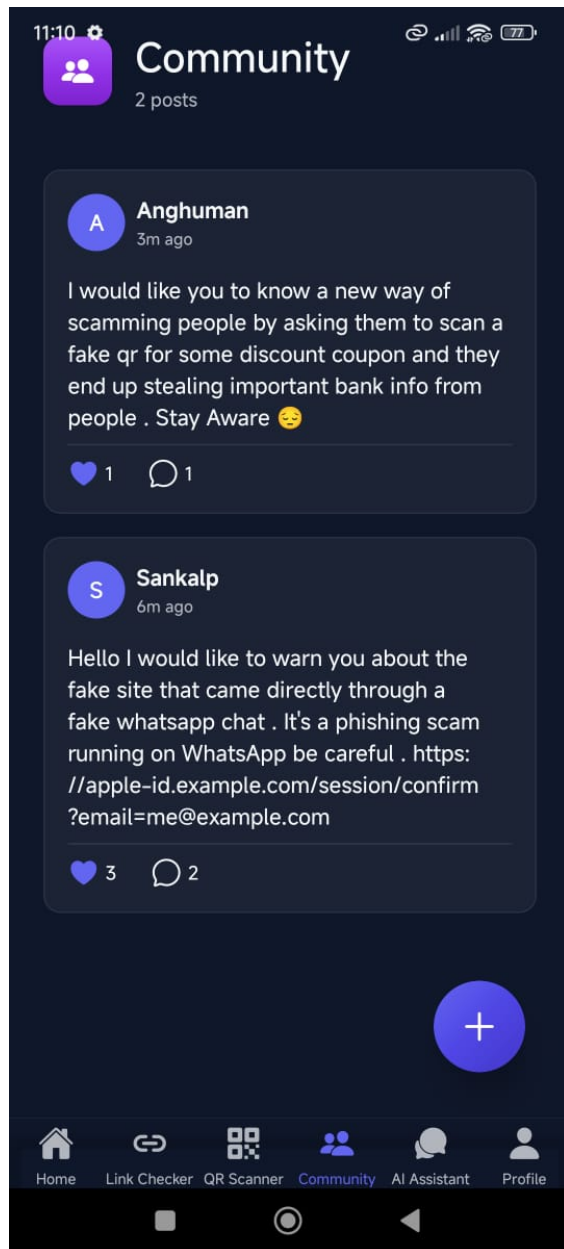


Figure 9: Community Feed

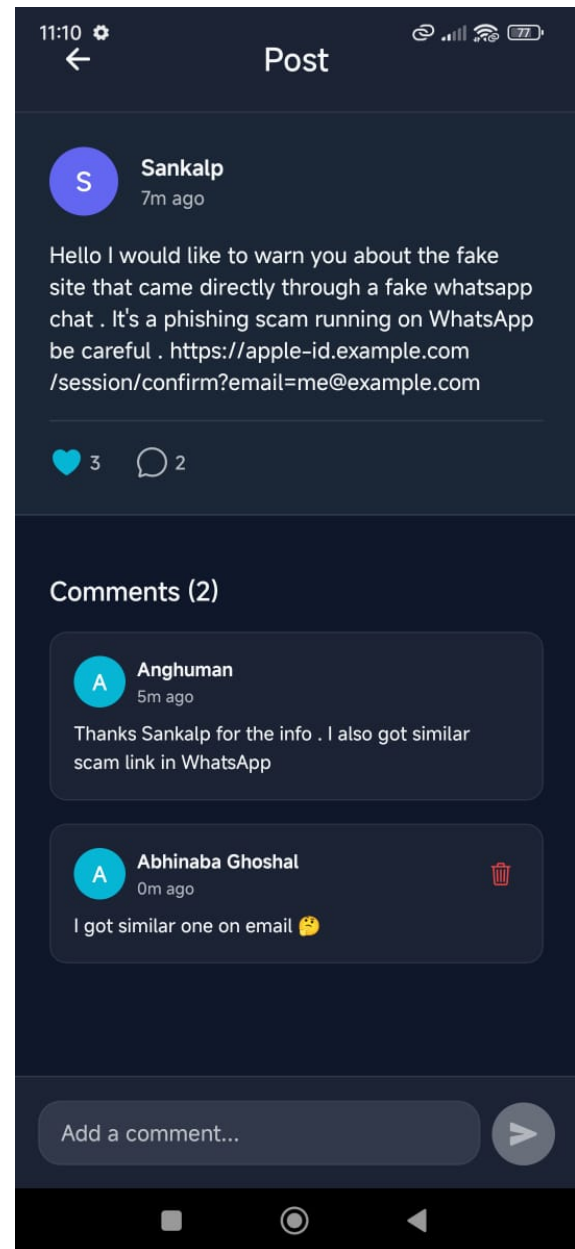


Figure 10: Post Details with Comments

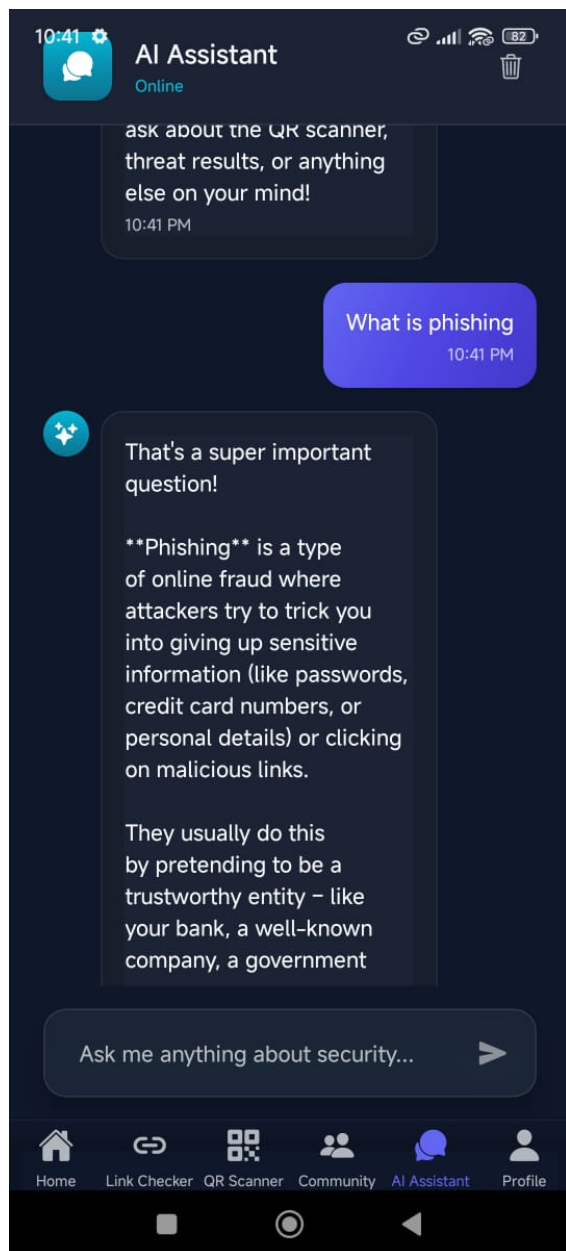


Figure 11: AI Security Assistant Chat

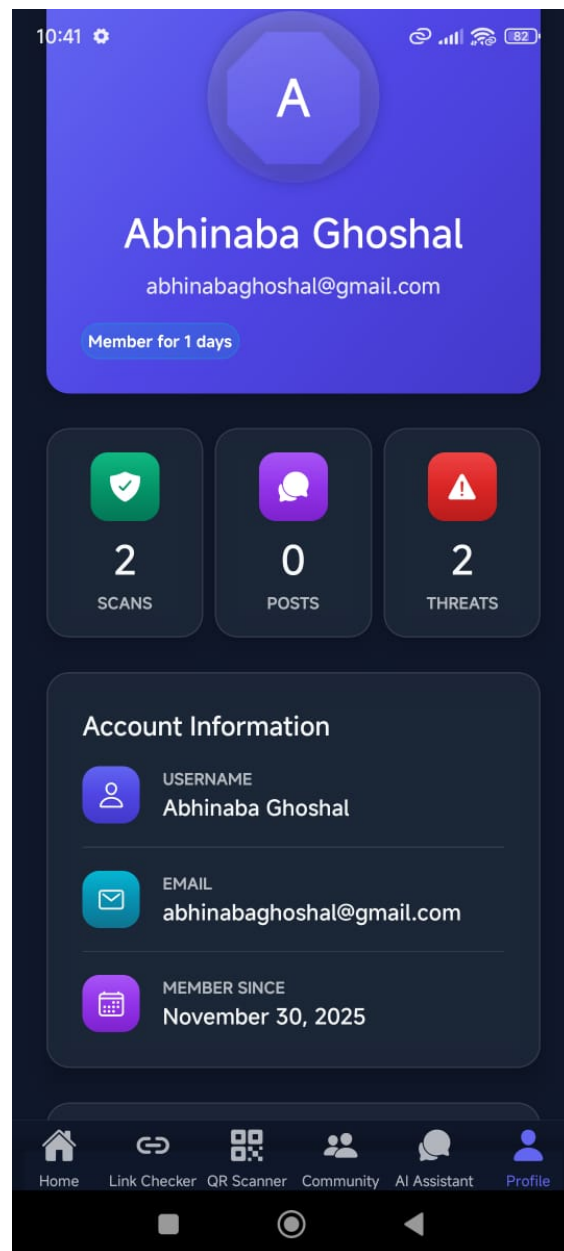


Figure 12: Profile &amp; Statistics

## 11 Appendix

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### 11.1 Repository Information

#### GitHub Repository

**Repository:** SecureScan

**Owner:** ShresthMantry

**Branch:** master

**Link:** <https://github.com/ShresthMantry/SecureScan>

**Structure:**

- /backend - Node.js Express API
- /frontend - React Native Expo app
- /ml-service - Python Flask ML service
- Documentation files in root directory